advance in this area since Euclid's time. Gauss' doctoral thesis of 1799 provided a rigorous proof of the Fundamental Theorem of Algebra, which had been stated first by Girard in 1629 and then proved imperfectly by d'Alembert (1746), and later by Euler (1749). The theorem (it asserts that a polynomial equation of degree n has exactly n complex roots) was always a favorite of Gauss', and he gave, in all, four distinct demonstrations of it. The publication of *Disquisitiones Arithmeticae* in 1801 at once placed Gauss in the front rank of mathematicians.

The most extraordinary achievement of Gauss was more in the realm of theoretical astronomy than of mathematics. On the opening night of the 19th century, January 1, 1801, the Italian astronomer Piazzi discovered the first of the so-called minor planets (planetoids or asteroids), later called Ceres. But after the course of this newly found body—visible only by telescope—passed the sun, neither Piazzi nor any other astronomer could locate it again. Piazzi's observations extended over a period of 41 days, during which the orbit swept out an angle of only nine degrees. From the scanty data available, Gauss was able to calculate the orbit of Ceres with amazing accuracy, and the elusive planet was rediscovered at the end of the year in almost exactly the position he had forecasted. This success brought Gauss worldwide fame, and led to his appointment as director of Göttingen Observatory.

By the middle of the 19th century, mathematics had grown into an enormous and unwieldy structure, divided into a large number of fields in which only the specialist knew his way. Gauss was the last complete mathematician, and it is no exaggeration to say that he was in some degree connected with nearly every aspect of the subject. His contemporaries regarded him as Princeps Mathematicorum (Prince of Mathematicians), on a par with Archimedes and Isaac Newton. This is revealed in a small incident: On being asked who was the greatest mathematician in Germany, Laplace answered, "Why, Pfaff." When the questioner indicated that he would have thought Gauss was, Laplace replied, "Pfaff is by far the greatest in Germany, but Gauss is the greatest in all Europe."

Although Gauss adorned every branch of mathematics, he always held number theory in high esteem and affection. He insisted that, "Mathematics is the Queen of the Sciences, and the theory of numbers is the Queen of Mathematics."

4.2 BASIC PROPERTIES OF CONGRUENCE

In the first chapter of *Disquisitiones Arithmeticae*, Gauss introduces the concept of congruence and the notation that makes it such a powerful technique (he explains that he was induced to adopt the symbol $\equiv$ because of the close analogy with algebraic equality). According to Gauss, "If a number n measures the difference between two numbers a and b, then a and b are said to be congruent with respect to n; if not, incongruent." Putting this into the form of a definition, we have Definition 4.1.

Definition 4.1. Let n be a fixed positive integer. Two integers a and b are said to be *congruent modulo* n, symbolized by

$$a \equiv b \pmod{n}$$

if n divides the difference $a - b$; that is, provided that $a - b = kn$ for some integer k.

To fix the idea, consider $n = 7$. It is routine to check that

$$3 \equiv 24 \pmod 7 \qquad -31 \equiv 11 \pmod 7 \qquad -15 \equiv -64 \pmod 7$$

because $3 - 24 = (-3)7$, $-31 - 11 = (-6)7$, and $-15 - (-64) = 7 \cdot 7$. When $n \nmid (a - b)$, we say that a is *incongruent to b modulo n*, and in this case we write $a \not\equiv b \pmod n$. For a simple example: $25 \not\equiv 12 \pmod 7$, because 7 fails to divide $25 - 12 = 13$.

It is to be noted that any two integers are congruent modulo 1, whereas two integers are congruent modulo 2 when they are both even or both odd. Inasmuch as congruence modulo 1 is not particularly interesting, the usual practice is to assume that $n > 1$.

Given an integer a, let q and r be its quotient and remainder upon division by n, so that

$$a = qn + r \qquad 0 \leq r < n$$

Then, by definition of congruence, $a \equiv r \pmod n$. Because there are n choices for r, we see that every integer is congruent modulo n to exactly one of the values $0, 1, 2, \ldots, n - 1$; in particular, $a \equiv 0 \pmod n$ if and only if $n \mid a$. The set of n integers $0, 1, 2, \ldots, n - 1$ is called the set of *least nonnegative residues modulo n*.

In general, a collection of n integers $a_1, a_2, \ldots, a_n$ is said to form a *complete set of residues* (or a *complete system of residues*) *modulo n* if every integer is congruent modulo n to one and only one of the a_k. To put it another way, $a_1, a_2, \ldots, a_n$ are congruent modulo n to $0, 1, 2, \ldots, n - 1$, taken in some order. For instance,

$$-12, -4, 11, 13, 22, 82, 91$$

constitute a complete set of residues modulo 7; here, we have

$$-12 \equiv 2 \quad -4 \equiv 3 \quad 11 \equiv 4 \quad 13 \equiv 6 \quad 22 \equiv 1 \quad 82 \equiv 5 \quad 91 \equiv 0$$

all modulo 7. An observation of some importance is that any n integers form a complete set of residues modulo n if and only if no two of the integers are congruent modulo n. We shall need this fact later.

Our first theorem provides a useful characterization of congruence modulo n in terms of remainders upon division by n.

Theorem 4.1. For arbitrary integers a and b, $a \equiv b \pmod n$ if and only if a and b leave the same nonnegative remainder when divided by n.

Proof. First take $a \equiv b \pmod n$, so that $a = b + kn$ for some integer k. Upon division by n, b leaves a certain remainder r; that is, $b = qn + r$, where $0 \leq r < n$. Therefore,

$$a = b + kn = (qn + r) + kn = (q + k)n + r$$

which indicates that a has the same remainder as b.

On the other hand, suppose we can write $a = q_1 n + r$ and $b = q_2 n + r$, with the same remainder r $(0 \leq r < n)$. Then

$$a - b = (q_1 n + r) - (q_2 n + r) = (q_1 - q_2)n$$

whence $n \mid a - b$. In the language of congruences, we have $a \equiv b \pmod n$.

Example 4.1. Because the integers -56 and -11 can be expressed in the form

$$-56 = (-7)9 + 7 \qquad -11 = (-2)9 + 7$$

with the same remainder 7, Theorem 4.1 tells us that $-56 \equiv -11 \pmod 9$. Going in the other direction, the congruence $-31 \equiv 11 \pmod 7$ implies that -31 and 11 have the same remainder when divided by 7; this is clear from the relations

$$-31 = (-5)7 + 4 \qquad 11 = 1 \cdot 7 + 4$$

Congruence may be viewed as a generalized form of equality, in the sense that its behavior with respect to addition and multiplication is reminiscent of ordinary equality. Some of the elementary properties of equality that carry over to congruences appear in the next theorem.

Theorem 4.2. Let $n > 1$ be fixed and a, b, c, d be arbitrary integers. Then the following properties hold:

(a) $a \equiv a \pmod n$.
(b) If $a \equiv b \pmod n$, then $b \equiv a \pmod n$.
(c) If $a \equiv b \pmod n$ and $b \equiv c \pmod n$, then $a \equiv c \pmod n$.
(d) If $a \equiv b \pmod n$ and $c \equiv d \pmod n$, then $a + c \equiv b + d \pmod n$ and $ac \equiv bd \pmod n$.
(e) If $a \equiv b \pmod n$, then $a + c \equiv b + c \pmod n$ and $ac \equiv bc \pmod n$.
(f) If $a \equiv b \pmod n$, then $a^k \equiv b^k \pmod n$ for any positive integer k.

Proof. For any integer a, we have $a - a = 0 \cdot n$, so that $a \equiv a \pmod n$. Now if $a \equiv b \pmod n$, then $a - b = kn$ for some integer k. Hence, $b - a = -(kn) = (-k)n$ and because $-k$ is an integer, this yields property (b).

Property (c) is slightly less obvious: Suppose that $a \equiv b \pmod n$ and also $b \equiv c \pmod n$. Then there exist integers h and k satisfying $a - b = hn$ and $b - c = kn$. It follows that

$$a - c = (a - b) + (b - c) = hn + kn = (h + k)n$$

which is $a \equiv c \pmod n$ in congruence notation.

In the same vein, if $a \equiv b \pmod n$ and $c \equiv d \pmod n$, then we are assured that $a - b = k_1 n$ and $c - d = k_2 n$ for some choice of k_1 and k_2. Adding these equations, we obtain

$$\begin{aligned}(a + c) - (b + d) &= (a - b) + (c - d) \\ &= k_1 n + k_2 n = (k_1 + k_2)n\end{aligned}$$

or, as a congruence statement, $a + c \equiv b + d \pmod n$. As regards the second assertion of property (d), note that

$$ac = (b + k_1 n)(d + k_2 n) = bd + (bk_2 + dk_1 + k_1 k_2 n)n$$

Because $bk_2 + dk_1 + k_1 k_2 n$ is an integer, this says that $ac - bd$ is divisible by n, whence $ac \equiv bd \pmod n$.

The proof of property (e) is covered by (d) and the fact that $c \equiv c \pmod n$. Finally, we obtain property (f) by making an induction argument. The statement certainly holds for $k = 1$, and we will assume it is true for some fixed k. From (d), we know

that $a \equiv b \pmod{n}$ and $a^k \equiv b^k \pmod{n}$ together imply that $aa^k \equiv bb^k \pmod{n}$, or equivalently $a^{k+1} \equiv b^{k+1} \pmod{n}$. This is the form the statement should take for $k+1$, and so the induction step is complete.

Before going further, we should illustrate that congruences can be a great help in carrying out certain types of computations.

Example 4.2. Let us endeavor to show that 41 divides $2^{20} - 1$. We begin by noting that $2^5 \equiv -9 \pmod{41}$, whence $(2^5)^4 \equiv (-9)^4 \pmod{41}$ by Theorem 4.2(f); in other words, $2^{20} \equiv 81 \cdot 81 \pmod{41}$. But $81 \equiv -1 \pmod{41}$, and so $81 \cdot 81 \equiv 1 \pmod{41}$. Using parts (b) and (e) of Theorem 4.2, we finally arrive at

$$2^{20} - 1 \equiv 81 \cdot 81 - 1 \equiv 1 - 1 \equiv 0 \pmod{41}$$

Thus, $41 \mid 2^{20} - 1$, as desired.

Example 4.3. For another example in the same spirit, suppose that we are asked to find the remainder obtained upon dividing the sum

$$1! + 2! + 3! + 4! + \cdots + 99! + 100!$$

by 12. Without the aid of congruences this would be an awesome calculation. The observation that starts us off is that $4! \equiv 24 \equiv 0 \pmod{12}$; thus, for $k \geq 4$,

$$k! \equiv 4! \cdot 5 \cdot 6 \cdots k \equiv 0 \cdot 5 \cdot 6 \cdots k \equiv 0 \pmod{12}$$

In this way, we find that

$$\begin{aligned} 1! + 2! + 3! + 4! + \cdots + 100! & \\ \equiv 1! + 2! + 3! + 0 + \cdots + 0 \equiv 9 \pmod{12} & \end{aligned}$$

Accordingly, the sum in question leaves a remainder of 9 when divided by 12.

In Theorem 4.1 we saw that if $a \equiv b \pmod{n}$, then $ca \equiv cb \pmod{n}$ for any integer c. The converse, however, fails to hold. As an example, perhaps as simple as any, note that $2 \cdot 4 \equiv 2 \cdot 1 \pmod{6}$, whereas $4 \not\equiv 1 \pmod{6}$. In brief: One cannot unrestrictedly cancel a common factor in the arithmetic of congruences.

With suitable precautions, cancellation can be allowed; one step in this direction, and an important one, is provided by the following theorem.

Theorem 4.3. If $ca \equiv cb \pmod{n}$, then $a \equiv b \pmod{n/d}$, where $d = \gcd(c, n)$.

Proof. By hypothesis, we can write

$$c(a - b) = ca - cb = kn$$

for some integer k. Knowing that $\gcd(c, n) = d$, there exist relatively prime integers r and s satisfying $c = dr$, $n = ds$. When these values are substituted in the displayed equation and the common factor d canceled, the net result is

$$r(a - b) = ks$$

Hence, $s \mid r(a - b)$ and $\gcd(r, s) = 1$. Euclid's lemma yields $s \mid a - b$, which may be recast as $a \equiv b \pmod{s}$; in other words, $a \equiv b \pmod{n/d}$.

Theorem 4.3 gets its maximum force when the requirement that $\gcd(c, n) = 1$ is added, for then the cancellation may be accomplished without a change in modulus.

Corollary 1. If $ca \equiv cb \pmod{n}$ and $\gcd(c, n) = 1$, then $a \equiv b \pmod{n}$.

We take a moment to record a special case of Corollary 1 that we shall have frequent occasion to use, namely, Corollary 2.

Corollary 2. If $ca \equiv cb \pmod{p}$ and $p \nmid c$, where p is a prime number, then $a \equiv b \pmod{p}$.

Proof. The conditions $p \nmid c$ and p a prime imply that $\gcd(c, p) = 1$.

Example 4.4. Consider the congruence $33 \equiv 15 \pmod{9}$ or, if one prefers, $3 \cdot 11 \equiv 3 \cdot 5 \pmod{9}$. Because $\gcd(3, 9) = 3$, Theorem 4.3 leads to the conclusion that $11 \equiv 5 \pmod{3}$. A further illustration is given by the congruence $-35 \equiv 45 \pmod{8}$, which is the same as $5 \cdot (-7) \equiv 5 \cdot 9 \pmod{8}$. The integers 5 and 8 being relatively prime, we may cancel the factor 5 to obtain a correct congruence $-7 \equiv 9 \pmod{8}$.

Let us call attention to the fact that, in Theorem 4.3, it is unnecessary to stipulate that $c \not\equiv 0 \pmod{n}$. Indeed, if $c \equiv 0 \pmod{n}$, then $\gcd(c, n) = n$ and the conclusion of the theorem would state that $a \equiv b \pmod{1}$; but, as we remarked earlier, this holds trivially for all integers a and b.

There is another curious situation that can arise with congruences: The product of two integers, neither of which is congruent to zero, may turn out to be congruent to zero. For instance, $4 \cdot 3 \equiv 0 \pmod{12}$, but $4 \not\equiv 0 \pmod{12}$ and $3 \not\equiv 0 \pmod{12}$. It is a simple matter to show that if $ab \equiv 0 \pmod{n}$ and $\gcd(a, n) = 1$, then $b \equiv 0 \pmod{n}$: Corollary 1 permits us legitimately to cancel the factor a from both sides of the congruence $ab \equiv a \cdot 0 \pmod{n}$. A variation on this is that when $ab \equiv 0 \pmod{p}$, with p a prime, then either $a \equiv 0 \pmod{p}$ or $b \equiv 0 \pmod{p}$.

PROBLEMS 4.2

1. Prove each of the following assertions:
   (a) If $a \equiv b \pmod{n}$ and $m \mid n$, then $a \equiv b \pmod{m}$.
   (b) If $a \equiv b \pmod{n}$ and $c > 0$, then $ca \equiv cb \pmod{cn}$.
   (c) If $a \equiv b \pmod{n}$ and the integers a, b, n are all divisible by $d > 0$, then $a/d \equiv b/d \pmod{n/d}$.
2. Give an example to show that $a^2 \equiv b^2 \pmod{n}$ need not imply that $a \equiv b \pmod{n}$.
3. If $a \equiv b \pmod{n}$, prove that $\gcd(a, n) = \gcd(b, n)$.
4. (a) Find the remainders when 2^{50} and 41^{65} are divided by 7.
   (b) What is the remainder when the following sum is divided by 4?

$$1^5 + 2^5 + 3^5 + \cdots + 99^5 + 100^5$$

5. Prove that the integer $53^{103} + 103^{53}$ is divisible by 39, and that $111^{333} + 333^{111}$ is divisible by 7.

6. For $n \geq 1$, use congruence theory to establish each of the following divisibility statements:
(a) $7 \mid 5^{2n} + 3 \cdot 2^{5n-2}$.
(b) $13 \mid 3^{n+2} + 4^{2n+1}$.
(c) $27 \mid 2^{5n+1} + 5^{n+2}$.
(d) $43 \mid 6^{n+2} + 7^{2n+1}$.

7. For $n \geq 1$, show that

$$(-13)^{n+1} \equiv (-13)^n + (-13)^{n-1} \pmod{181}$$

[*Hint:* Notice that $(-13)^2 \equiv -13 + 1 \pmod{181}$; use induction on n.]

8. Prove the assertions below:
(a) If a is an odd integer, then $a^2 \equiv 1 \pmod 8$.
(b) For any integer a, $a^3 \equiv 0, 1$, or $6 \pmod 7$.
(c) For any integer a, $a^4 \equiv 0$ or $1 \pmod 5$.
(d) If the integer a is not divisible by 2 or 3, then $a^2 \equiv 1 \pmod{24}$.

9. If p is a prime satisfying $n < p < 2n$, show that

$$\binom{2n}{n} \equiv 0 \pmod p$$

10. If $a_1, a_2, \ldots, a_n$ is a complete set of residues modulo n and $\gcd(a, n) = 1$, prove that $aa_1, aa_2, \ldots, aa_n$ is also a complete set of residues modulo n.
[*Hint:* It suffices to show that the numbers in question are incongruent modulo n.]

11. Verify that $0, 1, 2, 2^2, 2^3, \ldots, 2^9$ form a complete set of residues modulo 11, but that $0, 1^2, 2^2, 3^2, \ldots, 10^2$ do not.

12. Prove the following statements:
(a) If $\gcd(a, n) = 1$, then the integers

$$c, c + a, c + 2a, c + 3a, \ldots, c + (n - 1)a$$

form a complete set of residues modulo n for any c.
(b) Any n consecutive integers form a complete set of residues modulo n.
[*Hint:* Use part (a).]
(c) The product of any set of n consecutive integers is divisible by n.

13. Verify that if $a \equiv b \pmod{n_1}$ and $a \equiv b \pmod{n_2}$, then $a \equiv b \pmod n$, where the integer $n = \operatorname{lcm}(n_1, n_2)$. Hence, whenever n_1 and n_2 are relatively prime, $a \equiv b \pmod{n_1 n_2}$.

14. Give an example to show that $a^k \equiv b^k \pmod n$ and $k \equiv j \pmod n$ need not imply that $a^j \equiv b^j \pmod n$.

15. Establish that if a is an odd integer, then for any $n \geq 1$

$$a^{2^n} \equiv 1 \pmod{2^{n+2}}$$

[*Hint:* Proceed by induction on n.]

16. Use the theory of congruences to verify that

$$89 \mid 2^{44} - 1 \qquad \text{and} \qquad 97 \mid 2^{48} - 1$$

17. Prove that whenever $ab \equiv cd \pmod n$ and $b \equiv d \pmod n$, with $\gcd(b, n) = 1$, then $a \equiv c \pmod n$.

18. If $a \equiv b \pmod{n_1}$ and $a \equiv c \pmod{n_2}$, prove that $b \equiv c \pmod n$, where the integer $n = \gcd(n_1, n_2)$.

4.4 LINEAR CONGRUENCES AND THE CHINESE REMAINDER THEOREM

This is a convenient place in our development of number theory at which to investigate the theory of linear congruences: An equation of the form $ax \equiv b \pmod{n}$ is called a *linear congruence*, and by a solution of such an equation we mean an integer x_0 for which $ax_0 \equiv b \pmod{n}$. By definition, $ax_0 \equiv b \pmod{n}$ if and only if $n \mid ax_0 - b$ or, what amounts to the same thing, if and only if $ax_0 - b = ny_0$ for some integer y_0. Thus, the problem of finding all integers that will satisfy the linear congruence $ax \equiv b \pmod{n}$ is identical with that of obtaining all solutions of the linear Diophantine equation $ax - ny = b$. This allows us to bring the results of Chapter 2 into play.

It is convenient to treat two solutions of $ax \equiv b \pmod{n}$ that are congruent modulo n as being "equal" even though they are not equal in the usual sense. For instance, $x = 3$ and $x = -9$ both satisfy the congruence $3x \equiv 9 \pmod{12}$; because $3 \equiv -9 \pmod{12}$, they are not counted as different solutions. In short: When we refer to the number of solutions of $ax \equiv b \pmod{n}$, we mean the number of incongruent integers satisfying this congruence.

With these remarks in mind, the principal result is easy to state.

Theorem 4.7. The linear congruence $ax \equiv b \pmod{n}$ has a solution if and only if $d \mid b$, where $d = \gcd(a, n)$. If $d \mid b$, then it has d mutually incongruent solutions modulo n.

Proof. We already have observed that the given congruence is equivalent to the linear Diophantine equation $ax - ny = b$. From Theorem 2.9, it is known that the latter equation can be solved if and only if $d \mid b$; moreover, if it is solvable and x_0, y_0 is one specific solution, then any other solution has the form

$$x = x_0 + \frac{n}{d}t \qquad y = y_0 + \frac{a}{d}t$$

for some choice of t.

Among the various integers satisfying the first of these formulas, consider those that occur when t takes on the successive values $t = 0, 1, 2, \ldots, d - 1$:

$$x_0, x_0 + \frac{n}{d}, x_0 + \frac{2n}{d}, \ldots, x_0 + \frac{(d-1)n}{d}$$

We claim that these integers are incongruent modulo n, and all other such integers x are congruent to some one of them. If it happened that

$$x_0 + \frac{n}{d}t_1 \equiv x_0 + \frac{n}{d}t_2 \pmod{n}$$

where $0 \leq t_1 < t_2 \leq d - 1$, then we would have

$$\frac{n}{d}t_1 \equiv \frac{n}{d}t_2 \pmod{n}$$

Now $\gcd(n/d, n) = n/d$, and therefore by Theorem 4.3 the factor n/d could be canceled to arrive at the congruence

$$t_1 \equiv t_2 \pmod{d}$$

which is to say that $d \mid t_2 - t_1$. But this is impossible in view of the inequality $0 < t_2 - t_1 < d$.

It remains to argue that any other solution $x_0 + (n/d)t$ is congruent modulo n to one of the d integers listed above. The Division Algorithm permits us to write t as $t = qd + r$, where $0 \leq r \leq d - 1$. Hence

$$\begin{aligned} x_0 + \frac{n}{d}t &= x_0 + \frac{n}{d}(qd + r) \\ &= x_0 + nq + \frac{n}{d}r \\ &\equiv x_0 + \frac{n}{d}r \pmod{n} \end{aligned}$$

with $x_0 + (n/d)r$ being one of our d selected solutions. This ends the proof.

The argument that we gave in Theorem 4.7 brings out a point worth stating explicitly: If x_0 is any solution of $ax \equiv b \pmod{n}$, then the $d = \gcd(a, n)$ incongruent solutions are given by

$$x_0, x_0 + \frac{n}{d}, x_0 + 2\left(\frac{n}{d}\right), \ldots, x_0 + (d-1)\left(\frac{n}{d}\right)$$

For the reader's convenience, let us also record the form Theorem 4.7 takes in the special case in which a and n are assumed to be relatively prime.

Corollary. If $\gcd(a, n) = 1$, then the linear congruence $ax \equiv b \pmod{n}$ has a unique solution modulo n.

Given relatively prime integers a and n, the congruence $ax \equiv 1 \pmod{n}$ has a unique solution. This solution is sometimes called the (multiplicative) inverse of a modulo n.

We now pause to look at two concrete examples.

Example 4.7. First consider the linear congruence $18x \equiv 30 \pmod{42}$. Because $\gcd(18, 42) = 6$ and 6 surely divides 30, Theorem 4.7 guarantees the existence of exactly six solutions, which are incongruent modulo 42. By inspection, one solution is found to be $x = 4$. Our analysis tells us that the six solutions are as follows:

$$x \equiv 4 + (42/6)t \equiv 4 + 7t \pmod{42} \qquad t = 0, 1, \ldots, 5$$

or, plainly enumerated,

$$x \equiv 4, 11, 18, 25, 32, 39 \pmod{42}$$

Example 4.8. Let us solve the linear congruence $9x \equiv 21 \pmod{30}$. At the outset, because $\gcd(9, 30) = 3$ and $3 \mid 21$, we know that there must be three incongruent solutions.

One way to find these solutions is to divide the given congruence through by 3, thereby replacing it by the equivalent congruence $3x \equiv 7 \pmod{10}$. The relative primeness of 3 and 10 implies that the latter congruence admits a unique solution modulo 10. Although it is not the most efficient method, we could test the integers

0, 1, 2, . . . ,9 in turn until the solution is obtained. A better way is this: Multiply both sides of the congruence $3x \equiv 7 \pmod{10}$ by 7 to get

$$21x \equiv 49 \pmod{10}$$

which reduces to $x \equiv 9 \pmod{10}$. (This simplification is no accident, for the multiples $0 \cdot 3,\ 1 \cdot 3,\ 2 \cdot 3, \ldots, 9 \cdot 3$ form a complete set of residues modulo 10; hence, one of them is necessarily congruent to 1 modulo 10.) But the original congruence was given modulo 30, so that its incongruent solutions are sought among the integers 0, 1, 2, . . . , 29. Taking $t = 0, 1, 2$, in the formula

$$x = 9 + 10t$$

we obtain 9, 19, 29, whence

$$x \equiv 9 \pmod{30} \qquad x \equiv 19 \pmod{30} \qquad x \equiv 29 \pmod{30}$$

are the required three solutions of $9x \equiv 21 \pmod{30}$.

A different approach to the problem is to use the method that is suggested in the proof of Theorem 4.7. Because the congruence $9x \equiv 21 \pmod{30}$ is equivalent to the linear Diophantine equation

$$9x - 30y = 21$$

we begin by expressing $3 = \gcd(9, 30)$ as a linear combination of 9 and 30. It is found, either by inspection or by using the Euclidean Algorithm, that $3 = 9(-3) + 30 \cdot 1$, so that

$$21 = 7 \cdot 3 = 9(-21) - 30(-7)$$

Thus, $x = -21$, $y = -7$ satisfy the Diophantine equation and, in consequence, all solutions of the congruence in question are to be found from the formula

$$x = -21 + (30/3)t = -21 + 10t$$

The integers $x = -21 + 10t$, where $t = 0, 1, 2$, are incongruent modulo 30 (but all are congruent modulo 10); thus, we end up with the incongruent solutions

$$x \equiv -21 \pmod{30} \qquad x \equiv -11 \pmod{30} \qquad x \equiv -1 \pmod{30}$$

or, if one prefers positive numbers, $x \equiv 9, 19, 29 \pmod{30}$.

Having considered a single linear congruence, it is natural to turn to the problem of solving a system of simultaneous linear congruences:

$$a_1x \equiv b_1 \pmod{m_1}, a_2x \equiv b_2 \pmod{m_2}, \ldots, a_rx \equiv b_r \pmod{m_r}$$

We shall assume that the moduli m_k are relatively prime in pairs. Evidently, the system will admit no solution unless each individual congruence is solvable; that is, unless $d_k \mid b_k$ for each k, where $d_k = \gcd(a_k, m_k)$. When these conditions are satisfied, the factor d_k can be canceled in the kth congruence to produce a new system having the same set of solutions as the original one:

$$a_1'x \equiv b_1' \pmod{n_1}, a_2'x \equiv b_2' \pmod{n_2}, \ldots, a_r'x \equiv b_r' \pmod{n_r}$$

where $n_k = m_k/d_k$ and $\gcd(n_i, n_j) = 1$ for $i \neq j$; in addition, $\gcd(a_i', n_i) = 1$. The solutions of the individual congruences assume the form

$$x \equiv c_1 \pmod{n_1}, x \equiv c_2 \pmod{n_2}, \ldots, x \equiv c_r \pmod{n_r}$$

Thus, the problem is reduced to one of finding a simultaneous solution of a system of congruences of this simpler type.

The kind of problem that can be solved by simultaneous congruences has a long history, appearing in the Chinese literature as early as the 1st century A.D. Sun-Tsu asked: Find a number that leaves the remainders 2, 3, 2 when divided by 3, 5, 7, respectively. (Such mathematical puzzles are by no means confined to a single cultural sphere; indeed, the same problem occurs in the *Introductio Arithmeticae* of the Greek mathematician Nicomachus, circa 100 A.D.) In honor of their early contributions, the rule for obtaining a solution usually goes by the name of the Chinese Remainder Theorem.

Theorem 4.8 Chinese Remainder Theorem. Let $n_1, n_2, \ldots, n_r$ be positive integers such that $\gcd(n_i, n_j) = 1$ for $i \neq j$. Then the system of linear congruences

$$\begin{aligned} x &\equiv a_1 \pmod{n_1} \\ x &\equiv a_2 \pmod{n_2} \\ &\vdots \\ x &\equiv a_r \pmod{n_r} \end{aligned}$$

has a simultaneous solution, which is unique modulo the integer $n_1 n_2 \cdots n_r$.

Proof. We start by forming the product $n = n_1 n_2 \cdots n_r$. For each $k = 1, 2, \ldots, r$, let

$$N_k = \frac{n}{n_k} = n_1 \cdots n_{k-1} n_{k+1} \cdots n_r$$

In words, N_k is the product of all the integers n_i with the factor n_k omitted. By hypothesis, the n_i are relatively prime in pairs, so that $\gcd(N_k, n_k) = 1$. According to the theory of a single linear congruence, it is therefore possible to solve the congruence $N_k x \equiv 1 \pmod{n_k}$; call the unique solution x_k. Our aim is to prove that the integer

$$\bar{x} = a_1 N_1 x_1 + a_2 N_2 x_2 + \cdots + a_r N_r x_r$$

is a simultaneous solution of the given system.

First, observe that $N_i \equiv 0 \pmod{n_k}$ for $i \neq k$, because $n_k \mid N_i$ in this case. The result is

$$\bar{x} = a_1 N_1 x_1 + \cdots + a_r N_r x_r \equiv a_k N_k x_k \pmod{n_k}$$

But the integer x_k was chosen to satisfy the congruence $N_k x \equiv 1 \pmod{n_k}$, which forces

$$\bar{x} \equiv a_k \cdot 1 \equiv a_k \pmod{n_k}$$

This shows that a solution to the given system of congruences exists.

As for the uniqueness assertion, suppose that x' is any other integer that satisfies these congruences. Then

$$\bar{x} \equiv a_k \equiv x' \pmod{n_k} \qquad k = 1, 2, \ldots, r$$

and so $n_k \mid \bar{x} - x'$ for each value of k. Because $\gcd(n_i, n_j) = 1$, Corollary 2 to Theorem 2.4 supplies us with the crucial point that $n_1 n_2 \cdots n_r \mid \bar{x} - x'$; hence $\bar{x} \equiv x' \pmod{n}$. With this, the Chinese Remainder Theorem is proven.

Example 4.9. The problem posed by Sun-Tsu corresponds to the system of three congruences

$$x \equiv 2 \pmod 3$$
$$x \equiv 3 \pmod 5$$
$$x \equiv 2 \pmod 7$$

In the notation of Theorem 4.8, we have $n = 3 \cdot 5 \cdot 7 = 105$ and

$$N_1 = \frac{n}{3} = 35 \qquad N_2 = \frac{n}{5} = 21 \qquad N_3 = \frac{n}{7} = 15$$

Now the linear congruences

$$35x \equiv 1 \pmod 3 \qquad 21x \equiv 1 \pmod 5 \qquad 15x \equiv 1 \pmod 7$$

are satisfied by $x_1 = 2$, $x_2 = 1$, $x_3 = 1$, respectively. Thus, a solution of the system is given by

$$x = 2 \cdot 35 \cdot 2 + 3 \cdot 21 \cdot 1 + 2 \cdot 15 \cdot 1 = 233$$

Modulo 105, we get the unique solution $x = 233 \equiv 23 \pmod{105}$.

Example 4.10. For a second illustration, let us solve the linear congruence

$$17x \equiv 9 \pmod{276}$$

Because $276 = 3 \cdot 4 \cdot 23$, this is equivalent to finding a solution for the system of congruences

$$\begin{array}{lll} 17x \equiv 9 \pmod 3 & \text{or} & x \equiv 0 \pmod 3 \\ 17x \equiv 9 \pmod 4 & & x \equiv 1 \pmod 4 \\ 17x \equiv 9 \pmod{23} & & 17x \equiv 9 \pmod{23} \end{array}$$

Note that if $x \equiv 0 \pmod 3$, then $x = 3k$ for any integer k. We substitute into the second congruence of the system and obtain

$$3k \equiv 1 \pmod 4$$

Multiplication of both sides of this congruence by 3 gives us

$$k \equiv 9k \equiv 3 \pmod 4$$

so that $k = 3 + 4j$, where j is an integer. Then

$$x = 3(3 + 4j) = 9 + 12j$$

For x to satisfy the last congruence, we must have

$$17(9 + 12j) \equiv 9 \pmod{23}$$

or $204j \equiv -144 \pmod{23}$, which reduces to $3j \equiv 6 \pmod{23}$; in consequence, $j \equiv 2 \pmod{23}$. This yields $j = 2 + 23t$, with t an integer, whence

$$x = 9 + 12(2 + 23t) = 33 + 276t$$

All in all, $x \equiv 33 \pmod{276}$ provides a solution to the system of congruences and, in turn, a solution to $17x \equiv 9 \pmod{276}$.

We should say a few words about linear congruences in two variables; that is, congruences of the form

$$ax + by \equiv c \pmod{n}$$

In analogy with Theorem 4.7, such a congruence has a solution if and only if $\gcd(a, b, n)$ divides c. The condition for solvability holds if either $\gcd(a, n) = 1$ or $\gcd(b, n) = 1$. Say $\gcd(a, n) = 1$. When the congruence is expressed as

$$ax \equiv c - by \pmod{n}$$

the corollary to Theorem 4.7 guarantees a unique solution x for each of the n incongruent values of y. Take as a simple illustration $7x + 4y \equiv 5 \pmod{12}$, that would be treated as $7x \equiv 5 - 4y \pmod{12}$. Substitution of $y \equiv 5 \pmod{12}$ gives $7x \equiv -15 \pmod{12}$; but this is equivalent to $-5x \equiv -15 \pmod{12}$ so that $x \equiv 3 \pmod{12}$. It follows that $x \equiv 3 \pmod{12}, y \equiv 5 \pmod{12}$ is one of the 12 incongruent solutions of $7x + 4y \equiv 5 \pmod{12}$. Another solution having the same value of x is $x \equiv 3 \pmod{12}, y \equiv 8 \pmod{12}$.

The focus of our concern here is how to solve a system of two linear congruences in two variables with the same modulus. The proof of the coming theorem adopts the familiar procedure of eliminating one of the unknowns.

Theorem 4.9. The system of linear congruences

$$ax + by \equiv r \pmod{n}$$
$$cx + dy \equiv s \pmod{n}$$

has a unique solution modulo n whenever $\gcd(ad - bc, n) = 1$.

Proof. Let us multiply the first congruence of the system by d, the second congruence by b, and subtract the lower result from the upper. These calculations yield

$$(ad - bc)x \equiv dr - bs \pmod{n} \tag{1}$$

The assumption $\gcd(ad - bc, n) = 1$ ensures that the congruence

$$(ad - bc)z \equiv 1 \pmod{n}$$

posseses a unique solution; denote the solution by t. When congruence (1) is multiplied by t, we obtain

$$x \equiv t(dr - bs) \pmod{n}$$

A value for y is found by a similar elimination process. That is, multiply the first congruence of the system by c, the second one by a, and subtract to end up with

$$(ad - bc)y \equiv as - cr \pmod{n} \tag{2}$$

Multiplication of this congruence by t leads to

$$y \equiv t(as - cr) \pmod{n}$$

A solution of the system is now established.

We close this section with an example illustrating Theorem 4.9.

Example 4.11. Consider the system

$$7x + 3y \equiv 10 \pmod{16}$$
$$2x + 5y \equiv 9 \pmod{16}$$

Because $\gcd(7 \cdot 5 - 2 \cdot 3, 16) = \gcd(29, 16) = 1$, a solution exists. It is obtained by the method developed in the proof of Theorem 4.9. Multiplying the first congruence by 5, the second one by 3, and subtracting, we arrive at

$$29x \equiv 5 \cdot 10 - 3 \cdot 9 \equiv 23 \pmod{16}$$

or, what is the same thing, $13x \equiv 7 \pmod{16}$. Multiplication of this congruence by 5 (noting that $5 \cdot 13 \equiv 1 \pmod{16}$) produces $x \equiv 35 \equiv 3 \pmod{16}$. When the variable x is eliminated from the system of congruences in a like manner, it is found that

$$29y \equiv 7 \cdot 9 - 2 \cdot 10 \equiv 43 \pmod{16}$$

But then $13y \equiv 11 \pmod{16}$, which upon multiplication by 5, results in $y \equiv 55 \equiv 7 \pmod{16}$. The unique solution of our system turns out to be

$$x \equiv 3 \pmod{16} \qquad y \equiv 7 \pmod{16}$$

PROBLEMS 4.4

1. Solve the following linear congruences:
   (a) $25x \equiv 15 \pmod{29}$.
   (b) $5x \equiv 2 \pmod{26}$.
   (c) $6x \equiv 15 \pmod{21}$.
   (d) $36x \equiv 8 \pmod{102}$.
   (e) $34x \equiv 60 \pmod{98}$.
   (f) $140x \equiv 133 \pmod{301}$.
   [*Hint:* $\gcd(140, 301) = 7$.]
2. Using congruences, solve the Diophantine equations below:
   (a) $4x + 51y = 9$.
   [*Hint:* $4x \equiv 9 \pmod{51}$ gives $x = 15 + 51t$, whereas $51y \equiv 9 \pmod{4}$ gives $y = 3 + 4s$. Find the relation between s and t.]
   (b) $12x + 25y = 331$.
   (c) $5x - 53y = 17$.
3. Find all solutions of the linear congruence $3x - 7y \equiv 11 \pmod{13}$.
4. Solve each of the following sets of simultaneous congruences:
   (a) $x \equiv 1 \pmod{3}$, $x \equiv 2 \pmod{5}$, $x \equiv 3 \pmod{7}$.
   (b) $x \equiv 5 \pmod{11}$, $x \equiv 14 \pmod{29}$, $x \equiv 15 \pmod{31}$.
   (c) $x \equiv 5 \pmod{6}$, $x \equiv 4 \pmod{11}$, $x \equiv 3 \pmod{17}$.
   (d) $2x \equiv 1 \pmod{5}$, $3x \equiv 9 \pmod{6}$, $4x \equiv 1 \pmod{7}$, $5x \equiv 9 \pmod{11}$.
5. Solve the linear congruence $17x \equiv 3 \pmod{2 \cdot 3 \cdot 5 \cdot 7}$ by solving the system

$$17x \equiv 3 \pmod{2} \qquad 17x \equiv 3 \pmod{3}$$
$$17x \equiv 3 \pmod{5} \qquad 17x \equiv 3 \pmod{7}$$

6. Find the smallest integer $a > 2$ such that

$$2 \mid a,\ 3 \mid a + 1,\ 4 \mid a + 2, 5 \mid a + 3, 6 \mid a + 4$$

7. (a) Obtain three consecutive integers, each having a square factor.
[*Hint:* Find an integer a such that $2^2 \mid a, 3^2 \mid a+1, 5^2 \mid a+2$.]
(b) Obtain three consecutive integers, the first of which is divisible by a square, the second by a cube, and the third by a fourth power.

8. (Brahmagupta, 7th century A.D.) When eggs in a basket are removed 2, 3, 4, 5, 6 at a time there remain, respectively, 1, 2, 3, 4, 5 eggs. When they are taken out 7 at a time, none are left over. Find the smallest number of eggs that could have been contained in the basket.

9. The basket-of-eggs problem is often phrased in the following form: One egg remains when the eggs are removed from the basket 2, 3, 4, 5, or 6 at a time; but, no eggs remain if they are removed 7 at a time. Find the smallest number of eggs that could have been in the basket.

10. (Ancient Chinese Problem.) A band of 17 pirates stole a sack of gold coins. When they tried to divide the fortune into equal portions, 3 coins remained. In the ensuing brawl over who should get the extra coins, one pirate was killed. The wealth was redistributed, but this time an equal division left 10 coins. Again an argument developed in which another pirate was killed. But now the total fortune was evenly distributed among the survivors. What was the least number of coins that could have been stolen?

11. Prove that the congruences

$$x \equiv a \pmod{n} \qquad \text{and} \qquad x \equiv b \pmod{m}$$

admit a simultaneous solution if and only if $\gcd(n, m) \mid a - b$; if a solution exists, confirm that it is unique modulo $\operatorname{lcm}(n, m)$.

12. Use Problem 11 to show that the following system does not possess a solution:

$$x \equiv 5 \pmod{6} \qquad \text{and} \qquad x \equiv 7 \pmod{15}$$

13. If $x \equiv a \pmod{n}$, prove that either $x \equiv a \pmod{2n}$ or $x \equiv a + n \pmod{2n}$.

14. A certain integer between 1 and 1200 leaves the remainders 1, 2, 6 when divided by 9, 11, 13, respectively. What is the integer?

15. (a) Find an integer having the remainders 1, 2, 5, 5 when divided by 2, 3, 6, 12, respectively. (Yih-hing, died 717).
(b) Find an integer having the remainders 2, 3, 4, 5 when divided by 3, 4, 5, 6, respectively. (Bhaskara, born 1114).
(c) Find an integer having the remainders 3, 11, 15 when divided by 10, 13, 17, respectively. (Regiomontanus, 1436–1476).

16. Let t_n denote the nth triangular number. For which values of n does t_n divide

$$t_1^2 + t_2^2 + \cdots + t_n^2$$

[*Hint:* Because $t_1^2 + t_2^2 + \cdots + t_n^2 = t_n(3n^3 + 12n^2 + 13n + 2)/30$, it suffices to determine those n satisfying $3n^3 + 12n^2 + 13n + 2 \equiv 0 \pmod{2 \cdot 3 \cdot 5}$.]

17. Find the solutions of the system of congruences:

$$3x + 4y \equiv 5 \pmod{13}$$
$$2x + 5y \equiv 7 \pmod{13}$$

18. Obtain the two incongruent solutions modulo 210 of the system

$$2x \equiv 3 \pmod{5}$$
$$4x \equiv 2 \pmod{6}$$
$$3x \equiv 2 \pmod{7}$$

19. Obtain the eight incongruent solutions of the linear congruence $3x + 4y \equiv 5 \pmod 8$

20. Find the solutions of each of the following systems of congruences:

(a) $5x + 3y \equiv 1 \pmod 7$

$3x + 2y \equiv 4 \pmod 7$.

(b) $7x + 3y \equiv 6 \pmod{11}$

$4x + 2y \equiv 9 \pmod{11}$.

(c) $11x + 5y \equiv 7 \pmod{20}$

$6x + 3y \equiv 8 \pmod{20}$.

Fermat preferred the pleasure he derived from mathematical research itself to any reputation that it might bring him; indeed, he published only one major manuscript during his lifetime and that just 5 years before his death, using the concealing initials M.P.E.A.S. Adamantly refusing to put his work in finished form, he thwarted several efforts by others to make the results available in print under his name. In partial compensation for his lack of interest in publication, Fermat carried on a voluminous correspondence with contemporary mathematicians. Most of what little we know about his investigations is found in the letters to friends with whom he exchanged problems and to whom he reported his successes. They did their best to publicize Fermat's talents by passing these letters from hand to hand or by making copies, which were dispatched over the Continent.

As his parliamentary duties demanded an ever greater portion of his time, Fermat was given to inserting notes in the margin of whatever book he happened to be using. Fermat's personal copy of the Bachet edition of Diophantus held in its margin many of his famous theorems in number theory. These were discovered by his son Samuel 5 years after Fermat's death. His son brought out a new edition of the *Arithmetica* incorporating Fermat's celebrated marginalia. Because there was little space available, Fermat's habit had been to jot down some result and omit all steps leading to the conclusion. Posterity has wished many times that the margins of the *Arithmetica* had been wider or that Fermat had been a little less secretive about his methods.

5.2 FERMAT'S LITTLE THEOREM AND PSEUDOPRIMES

The most significant of Fermat's correspondents in number theory was Bernhard Frénicle de Bessy (1605–1675), an official at the French mint who was renowned for his gift of manipulating large numbers. (Frénicle's facility in numerical calculation is revealed by the following incident: On hearing that Fermat had proposed the problem of finding cubes that when increased by their proper divisors become squares, as is the case with $7^3 + (1 + 7 + 7^2) = 20^2$, he immediately gave four different solutions, and supplied six more the next day.) Though in no way Fermat's equal as a mathematician, Frénicle alone among his contemporaries could challenge Fermat in number theory and Frénicle's challenges had the distinction of coaxing out of Fermat some of his carefully guarded secrets. One of the most striking is the theorem that states: If p is a prime and a is any integer not divisible by p, then p divides $a^{p-1} - 1$. Fermat communicated the result in a letter to Frénicle dated October 18, 1640, along with the comment, "I would send you the demonstration, if I did not fear its being too long." This theorem has since become known as "Fermat's Little Theorem," or just "Fermat's Theorem," to distinguish it from Fermat's "Great" or "Last Theorem," which is the subject of Chapter 12. Almost 100 years were to elapse before Euler published the first proof of the little theorem in 1736. Leibniz, however, seems not to have received his share of recognition, for he left an identical argument in an unpublished manuscript sometime before 1683.

We now proceed to a proof of Fermat's theorem.

Theorem 5.1 Fermat's theorem. Let p be a prime and suppose that $p \nmid a$. Then $a^{p-1} \equiv 1 \pmod{p}$.

Proof. We begin by considering the first $p-1$ positive multiples of a; that is, the integers

$$a, 2a, 3a, \ldots, (p-1)a$$

None of these numbers is congruent modulo p to any other, nor is any congruent to zero. Indeed, if it happened that

$$ra \equiv sa \pmod{p} \qquad 1 \leq r < s \leq p-1$$

then a could be canceled to give $r \equiv s \pmod{p}$, which is impossible. Therefore, the previous set of integers must be congruent modulo p to $1, 2, 3, \ldots, p-1$, taken in some order. Multiplying all these congruences together, we find that

$$a \cdot 2a \cdot 3a \cdots (p-1)a \equiv 1 \cdot 2 \cdot 3 \cdots (p-1) \pmod{p}$$

whence

$$a^{p-1}(p-1)! \equiv (p-1)! \pmod{p}$$

Once $(p-1)!$ is canceled from both sides of the preceding congruence (this is possible because since $p \nmid (p-1)!$), our line of reasoning culminates in the statement that $a^{p-1} \equiv 1 \pmod{p}$, which is Fermat's theorem.

This result can be stated in a slightly more general way in which the requirement that $p \nmid a$ is dropped.

Corollary. If p is a prime, then $a^p \equiv a \pmod{p}$ for any integer a.

Proof. When $p \mid a$, the statement obviously holds; for, in this setting, $a^p \equiv 0 \equiv a \pmod{p}$. If $p \nmid a$, then according to Fermat's theorem, we have $a^{p-1} \equiv 1 \pmod{p}$. When this congruence is multiplied by a, the conclusion $a^p \equiv a \pmod{p}$ follows.

There is a different proof of the fact that $a^p \equiv a \pmod{p}$, involving induction on a. If $a = 1$, the assertion is that $1^p \equiv 1 \pmod{p}$, which clearly is true, as is the case $a = 0$. Assuming that the result holds for a, we must confirm its validity for $a+1$. In light of the binomial theorem,

$$(a+1)^p = a^p + \binom{p}{1} a^{p-1} + \cdots + \binom{p}{k} a^{p-k} + \cdots + \binom{p}{p-1} a + 1$$

where the coefficient $\binom{p}{k}$ is given by

$$\binom{p}{k} = \frac{p!}{k!(p-k)!} = \frac{p(p-1)\cdots(p-k+1)}{1 \cdot 2 \cdot 3 \cdots k}$$

Our argument hinges on the observation that $\binom{p}{k} \equiv 0 \pmod{p}$ for $1 \leq k \leq p-1$. To see this, note that

$$k! \binom{p}{k} = p(p-1)\cdots(p-k+1) \equiv 0 \pmod{p}$$

by virtue of which $p \mid k!$ or $p \mid \binom{p}{k}$. But $p \mid k!$ implies that $p \mid j$ for some j satisfying $1 \leq j \leq k \leq p-1$, an absurdity. Therefore, $p \mid \binom{p}{k}$ or, converting to a congruence statement,

$$\binom{p}{k} \equiv 0 \pmod{p}$$

The point we wish to make is that

$$(a+1)^p \equiv a^p + 1 \equiv a + 1 \pmod{p}$$

where the rightmost congruence uses our inductive assumption. Thus, the desired conclusion holds for $a+1$ and, in consequence, for all $a \geq 0$. If a happens to be a negative integer, there is no problem: because $a \equiv r \pmod{p}$ for some r, where $0 \leq r \leq p-1$, we get $a^p \equiv r^p \equiv r \equiv a \pmod{p}$.

Fermat's theorem has many applications and is central to much of what is done in number theory. In the least, it can be a labor-saving device in certain calculations. If asked to verify that $5^{38} \equiv 4 \pmod{11}$, for instance, we take the congruence $5^{10} \equiv 1 \pmod{11}$ as our starting point. Knowing this,

$$\begin{aligned} 5^{38} = 5^{10 \cdot 3 + 8} &= (5^{10})^3 (5^2)^4 \\ &\equiv 1^3 \cdot 3^4 \equiv 81 \equiv 4 \pmod{11} \end{aligned}$$

as desired.

Another use of Fermat's theorem is as a tool in testing the primality of a given integer n. If it could be shown that the congruence

$$a^n \equiv a \pmod{n}$$

fails to hold for some choice of a, then n is necessarily composite. As an example of this approach, let us look at $n = 117$. The computation is kept under control by selecting a small integer for a, say, $a = 2$. Because 2^{117} may be written as

$$2^{117} = 2^{7 \cdot 16 + 5} = (2^7)^{16} 2^5$$

and $2^7 = 128 \equiv 11 \pmod{117}$, we have

$$2^{117} \equiv 11^{16} \cdot 2^5 \equiv (121)^8 2^5 \equiv 4^8 \cdot 2^5 \equiv 2^{21} \pmod{117}$$

But $2^{21} = (2^7)^3$, which leads to

$$2^{21} \equiv 11^3 \equiv 121 \cdot 11 \equiv 4 \cdot 11 \equiv 44 \pmod{117}$$

Combining these congruences, we finally obtain

$$2^{117} \equiv 44 \not\equiv 2 \pmod{117}$$

so that 117 must be composite; actually, $117 = 13 \cdot 9$.

It might be worthwhile to give an example illustrating the failure of the converse of Fermat's theorem to hold, in other words, to show that if $a^{n-1} \equiv 1 \pmod{n}$ for some integer a, then n need not be prime. As a prelude we require a technical lemma.

Lemma. If p and q are distinct primes with $a^p \equiv a \pmod{q}$ and $a^q \equiv a \pmod{p}$, then $a^{pq} \equiv a \pmod{pq}$.

Proof. The last corollary tells us that $(a^q)^p \equiv a^q \pmod{p}$, whereas $a^q \equiv a \pmod{p}$ holds by hypothesis. Combining these congruences, we obtain $a^{pq} \equiv a \pmod{p}$ or, in different terms, $p \mid a^{pq} - a$. In an entirely similar manner, $q \mid a^{pq} - a$. Corollary 2 to Theorem 2.4 now yields $pq \mid a^{pq} - a$, which can be recast as $a^{pq} \equiv a \pmod{pq}$.

Our contention is that $2^{340} \equiv 1 \pmod{341}$, where $341 = 11 \cdot 31$. In working toward this end, notice that $2^{10} = 1024 = 31 \cdot 33 + 1$. Thus,

$$2^{11} = 2 \cdot 2^{10} \equiv 2 \cdot 1 \equiv 2 \pmod{31}$$

and

$$2^{31} = 2(2^{10})^3 \equiv 2 \cdot 1^3 \equiv 2 \pmod{11}$$

Exploiting the lemma,

$$2^{11 \cdot 31} \equiv 2 \pmod{11 \cdot 31}$$

or $2^{341} \equiv 2 \pmod{341}$. After canceling a factor of 2, we pass to

$$2^{340} \equiv 1 \pmod{341}$$

so that the converse to Fermat's theorem is false.

The historical interest in numbers of the form $2^n - 2$ resides in the claim made by Chinese mathematicians over 25 centuries ago that n is prime if and only if $n \mid 2^n - 2$ (in point of fact, this criterion is reliable for all integers $n \leq 340$). Our example, where $341 \mid 2^{341} - 2$, although $341 = 11 \cdot 31$, lays the conjecture to rest; this was discovered in the year 1819. The situation in which $n \mid 2^n - 2$ occurs often enough to merit a name, though: A composite integer n is called *pseudoprime* whenever $n \mid 2^n - 2$. It can be shown that there are infinitely many pseudoprimes, the smallest four being 341, 561, 645, and 1105.

Theorem 5.2 allows us to construct an increasing sequence of pseudoprimes.

Theorem 5.2. If n is an odd pseudoprime, then $M_n = 2^n - 1$ is a larger one.

Proof. Because n is a composite number, we can write $n = rs$, with $1 < r \leq s < n$. Then, according to Problem 21, Section 2.3, $2^r - 1 \mid 2^n - 1$, or equivalently $2^r - 1 \mid M_n$, making M_n composite. By our hypotheses, $2^n \equiv 2 \pmod{n}$; hence $2^n - 2 = kn$ for some integer k. It follows that

$$2^{M_n - 1} = 2^{2^n - 2} = 2^{kn}$$

This yields

$$\begin{aligned} 2^{M_n - 1} - 1 &= 2^{kn} - 1 \\ &= (2^n - 1)(2^{n(k-1)} + 2^{n(k-2)} + \cdots + 2^n + 1) \\ &= M_n(2^{n(k-1)} + 2^{n(k-2)} + \cdots + 2^n + 1) \\ &\equiv 0 \pmod{M_n} \end{aligned}$$

We see immediately that $2^{M_n} - 2 \equiv 0 \pmod{M_n}$, in light of which M_n is a pseudoprime.

More generally, a composite integer n for which $a^n \equiv a \pmod{n}$ is called a *pseudoprime to the base a*. (When $a = 2$, n is simply said to be a pseudoprime.) For instance, 91 is the smallest pseudoprime to base 3, whereas 217 is the smallest such to base 5. It has been proved (1903) that there are infinitely many pseudoprimes to any given base.

These "prime imposters" are much rarer than are actual primes. Indeed, there are only 245 pseudoprimes smaller than one million, in comparison with 78498 primes. The first example of an even pseudoprime, namely, the number

$$161038 = 2 \cdot 73 \cdot 1103$$

was found in 1950.

There exist composite numbers n that are pseudoprimes to every base a; that is, $a^n \equiv a \pmod{n}$ for all integers a. The least such is 561. These exceptional numbers are called *absolute pseudoprimes* or *Carmichael numbers*, for R. D. Carmichael, who was the first to notice their existence. In his first paper on the subject, published in 1910, Carmichael indicated four absolute pseudoprimes including the well-known $561 = 3 \cdot 11 \cdot 17$; the others are $1105 = 5 \cdot 13 \cdot 17, 2821 = 7 \cdot 13 \cdot 31$, and $15841 = 7 \cdot 31 \cdot 73$. Two years later he presented 11 more having three prime factors and discovered one absolute pseudoprime with four factors, specifically, $16046641 = 13 \cdot 37 \cdot 73 \cdot 457$. The largest number of this kind known to date is the product of 1101518 distinct odd primes: It has 16142049 digits.

To see that $561 = 3 \cdot 11 \cdot 17$ must be an absolute pseudoprime, notice that $\gcd(a, 561) = 1$ gives

$$\gcd(a, 3) = \gcd(a, 11) = \gcd(a, 17) = 1$$

An application of Fermat's theorem leads to the congruences

$$a^2 \equiv 1 \pmod{3} \qquad a^{10} \equiv 1 \pmod{11} \qquad a^{16} \equiv 1 \pmod{17}$$

and, in turn, to

$$\begin{aligned} a^{560} &\equiv (a^2)^{280} \equiv 1 \pmod{3} \\ a^{560} &\equiv (a^{10})^{56} \equiv 1 \pmod{11} \\ a^{560} &\equiv (a^{16})^{35} \equiv 1 \pmod{17} \end{aligned}$$

These give rise to the single congruence $a^{560} \equiv 1 \pmod{561}$, where $\gcd(a, 561) = 1$. But then $a^{561} \equiv a \pmod{561}$ for all a, showing 561 to be an absolute pseudoprime.

Any absolute pseudoprime is square-free. This is easy to prove. Suppose that $a^n \equiv a \pmod{n}$ for every integer a, but $k^2 \mid n$ for some $k > 1$. If we let $a = k$, then $k^n \equiv k \pmod{n}$. Because $k^2 \mid n$, this last congruence holds modulo k^2; that is, $k \equiv k^n \equiv 0 \pmod{k^2}$, whence $k^2 \mid k$, which is impossible. Thus, n must be square-free.

Next we present a theorem that furnishes a means for producing absolute pseudoprimes.

Theorem 5.3. Let n be a composite square-free integer, say, $n = p_1 p_2 \cdots p_r$, where the p_i are distinct primes. If $p_i - 1 \mid n - 1$ for $i = 1, 2, \ldots, r$, then n is an absolute pseudoprime.

Proof. Suppose that a is an integer satisfying $\gcd(a, n) = 1$, so that $\gcd(a, p_i) = 1$ for each i. Then Fermat's theorem yields $p_i \mid a^{p_i - 1} - 1$. From the divisibility hypothesis $p_i - 1 \mid n - 1$, we have $p_i \mid a^{n-1} - 1$, and therefore $p_i \mid a^n - a$ for all a and $i = 1, 2, \ldots, r$. As a result of Corollary 2 to Theorem 2.4, we end up with $n \mid a^n - a$, which makes n an absolute pseudoprime.

Examples of integers that satisfy the conditions of Theorem 5.3 are

$$1729 = 7 \cdot 13 \cdot 19 \qquad 6601 = 7 \cdot 23 \cdot 41 \qquad 10585 = 5 \cdot 29 \cdot 73$$

It was proven in 1994 that infinitely many absolute pseudoprimes exist, but that they are fairly rare. There are just 43 of them less than one million, and 105212 less than 10^{15}.

PROBLEMS 5.2

1. Use Fermat's theorem to verify that 17 divides $11^{104} + 1$.

2. (a) If $\gcd(a, 35) = 1$, show that $a^{12} \equiv 1 \pmod{35}$.
[*Hint:* From Fermat's theorem $a^6 \equiv 1 \pmod 7$ and $a^4 \equiv 1 \pmod 5$.]
(b) If $\gcd(a, 42) = 1$, show that $168 = 3 \cdot 7 \cdot 8$ divides $a^6 - 1$.
(c) If $\gcd(a, 133) = \gcd(b, 133) = 1$, show that $133 \mid a^{18} - b^{18}$.

3. From Fermat's theorem deduce that, for any integer $n \geq 0$, $13 \mid 11^{12n+6} + 1$.

4. Derive each of the following congruences:
(a) $a^{21} \equiv a \pmod{15}$ for all a.
[*Hint:* By Fermat's theorem, $a^5 \equiv a \pmod 5$.]
(b) $a^7 \equiv a \pmod{42}$ for all a.
(c) $a^{13} \equiv a \pmod{3 \cdot 7 \cdot 13}$ for all a.
(d) $a^9 \equiv a \pmod{30}$ for all a.

5. If $\gcd(a, 30) = 1$, show that 60 divides $a^4 + 59$.

6. (a) Find the units digit of 3^{100} by the use of Fermat's theorem.
(b) For any integer a, verify that a^5 and a have the same units digit.

7. If $7 \nmid a$, prove that either $a^3 + 1$ or $a^3 - 1$ is divisible by 7.

8. The three most recent appearances of Halley's comet were in the years 1835, 1910, and 1986; the next occurrence will be in 2061. Prove that

$$1835^{1910} + 1986^{2061} \equiv 0 \pmod 7$$

9. (a) Let p be a prime and $\gcd(a, p) = 1$. Use Fermat's theorem to verify that $x \equiv a^{p-2}b \pmod p$ is a solution of the linear congruence $ax \equiv b \pmod p$.
(b) By applying part (a), solve the congruences $2x \equiv 1 \pmod{31}$, $6x \equiv 5 \pmod{11}$, and $3x \equiv 17 \pmod{29}$.

10. Assuming that a and b are integers not divisible by the prime p, establish the following:
(a) If $a^p \equiv b^p \pmod p$, then $a \equiv b \pmod p$.
(b) If $a^p \equiv b^p \pmod p$, then $a^p \equiv b^p \pmod{p^2}$.
[*Hint:* By (a), $a = b + pk$ for some k, so that $a^p - b^p = (b + pk)^p - b^p$; now show that p^2 divides the latter expression.]

11. Employ Fermat's theorem to prove that, if p is an odd prime, then
(a) $1^{p-1} + 2^{p-1} + 3^{p-1} + \cdots + (p-1)^{p-1} \equiv -1 \pmod p$.
(b) $1^p + 2^p + 3^p + \cdots + (p-1)^p \equiv 0 \pmod p$.
[*Hint:* Recall the identity $1 + 2 + 3 + \cdots + (p-1) = p(p-1)/2$.]

12. Prove that if p is an odd prime and k is an integer satisfying $1 \leq k \leq p-1$, then the binomial coefficient

$$\binom{p-1}{k} \equiv (-1)^k \pmod{p}$$

13. Assume that p and q are distinct odd primes such that $p-1 \mid q-1$. If $\gcd(a, pq) = 1$, show that $a^{q-1} \equiv 1 \pmod{pq}$.

14. If p and q are distinct primes, prove that

$$p^{q-1} + q^{p-1} \equiv 1 \pmod{pq}$$

15. Establish the statements below:

(a) If the number $M_p = 2^p - 1$ is composite, where p is a prime, then M_p is a pseudoprime.

(b) Every composite number $F_n = 2^{2^n} + 1$ is a pseudoprime ($n = 0, 1, 2, \ldots$).
[*Hint:* By Problem 21, Section 2.3, $2^{n+1} \mid 2^{2^n}$ implies that $2^{2^{n+1}} - 1 \mid 2^{F_n - 1} - 1$; but $F_n \mid 2^{2^{n+1}} - 1$.]

16. Confirm that the following integers are absolute pseudoprimes:

(a) $1105 = 5 \cdot 13 \cdot 17$.

(b) $2821 = 7 \cdot 13 \cdot 31$.

(c) $2465 = 5 \cdot 17 \cdot 29$.

17. Show that the smallest pseudoprime 341 is not an absolute pseudoprime by showing that $11^{341} \not\equiv 11 \pmod{341}$.
[*Hint:* $31 \nmid 11^{341} - 11$.]

18. (a) When $n = 2p$, where p is an odd prime, prove that $a^{n-1} \equiv a \pmod{n}$ for any integer a.

(b) For $n = 195 = 3 \cdot 5 \cdot 13$, verify that $a^{n-2} \equiv a \pmod{n}$ for any integer a.

19. Prove that any integer of the form

$$n = (6k+1)(12k+1)(18k+1)$$

is an absolute pseudoprime if all three factors are prime; hence, $1729 = 7 \cdot 13 \cdot 19$ is an absolute pseudoprime.

20. Show that $561 \mid 2^{561} - 2$ and $561 \mid 3^{561} - 3$. It is an unanswered question whether there exist infinitely many composite numbers n with the property that $n \mid 2^n - 2$ and $n \mid 3^n - 3$.

21. Establish the congruence

$$2222^{5555} + 5555^{2222} \equiv 0 \pmod{7}$$

[*Hint:* First evaluate 1111 modulo 7.]

5.3 WILSON'S THEOREM

We now turn to another milestone in the development of number theory. In his *Meditationes Algebraicae* of 1770, the English mathematician Edward Waring (1734–1798) announced several new theorems. Foremost among these is an interesting property of primes reported to him by one of his former students, a certain John Wilson. The property is the following: If p is a prime number, then p divides $(p-1)! + 1$. Wilson appears to have guessed this on the basis of numerical computations; at any rate, neither he nor Waring knew how to prove it. Confessing his inability to supply a demonstration, Waring added, "Theorems of this kind will be

very hard to prove, because of the absence of a notation to express prime numbers." (Reading the passage, Gauss uttered his telling comment on "notationes versus notiones," implying that in questions of this nature it was the notion that really mattered, not the notation.) Despite Waring's pessimistic forecast, soon afterward Lagrange (1771) gave a proof of what in literature is called "Wilson's theorem" and observed that the converse also holds. Perhaps it would be more just to name the theorem after Leibniz, for there is evidence that he was aware of the result almost a century earlier, but published nothing on the subject.

Now we give a proof of Wilson's theorem.

Theorem 5.4 Wilson. If p is a prime, then $(p-1)! \equiv -1 \pmod{p}$.

Proof. Dismissing the cases $p = 2$ and $p = 3$ as being evident, let us take $p > 3$. Suppose that a is any one of the $p - 1$ positive integers

$$1, 2, 3, \ldots, p-1$$

and consider the linear congruence $ax \equiv 1 \pmod{p}$. Then $\gcd(a, p) = 1$. By Theorem 4.7, this congruence admits a unique solution modulo p; hence, there is a unique integer a', with $1 \leq a' \leq p-1$, satisfying $aa' \equiv 1 \pmod{p}$.

Because p is prime, $a = a'$ if and only if $a = 1$ or $a = p - 1$. Indeed, the congruence $a^2 \equiv 1 \pmod{p}$ is equivalent to $(a-1)\cdot(a+1) \equiv 0 \pmod{p}$. Therefore, either $a - 1 \equiv 0 \pmod{p}$, in which case $a = 1$, or $a + 1 \equiv 0 \pmod{p}$, in which case $a = p - 1$.

If we omit the numbers 1 and $p - 1$, the effect is to group the remaining integers $2, 3, \ldots, p-2$ into pairs a, a', where $a \neq a'$, such that their product $aa' \equiv 1 \pmod{p}$. When these $(p-3)/2$ congruences are multiplied together and the factors rearranged, we get

$$2 \cdot 3 \cdots (p-2) \equiv 1 \pmod{p}$$

or rather

$$(p-2)! \equiv 1 \pmod{p}$$

Now multiply by $p - 1$ to obtain the congruence

$$(p-1)! \equiv p - 1 \equiv -1 \pmod{p}$$

as was to be proved.

Example 5.1. A concrete example should help to clarify the proof of Wilson's theorem. Specifically, let us take $p = 13$. It is possible to divide the integers $2, 3, \ldots, 11$ into $(p-3)/2 = 5$ pairs, each product of which is congruent to 1 modulo 13. To write these congruences out explicitly:

$$\begin{aligned} 2 \cdot 7 &\equiv 1 \pmod{13} \\ 3 \cdot 9 &\equiv 1 \pmod{13} \\ 4 \cdot 10 &\equiv 1 \pmod{13} \\ 5 \cdot 8 &\equiv 1 \pmod{13} \\ 6 \cdot 11 &\equiv 1 \pmod{13} \end{aligned}$$

Multiplying these congruences gives the result

$$11! = (2 \cdot 7)(3 \cdot 9)(4 \cdot 10)(5 \cdot 8)(6 \cdot 11) \equiv 1 \pmod{13}$$

and so

$$12! \equiv 12 \equiv -1 \pmod{13}$$

Thus, $(p-1)! \equiv -1 \pmod{p}$, with $p = 13$.

The converse of Wilson's theorem is also true. If $(n-1)! \equiv -1 \pmod{n}$, then n must be prime. For, if n is not a prime, then n has a divisor d with $1 < d < n$. Furthermore, because $d \leq n-1$, d occurs as one of the factors in $(n-1)!$, whence $d \mid (n-1)!$. Now we are assuming that $n \mid (n-1)! + 1$, and so $d \mid (n-1)! + 1$, too. The conclusion is that $d \mid 1$, which is nonsense.

Taken together, Wilson's theorem and its converse provide a necessary and sufficient condition for determining primality; namely, an integer $n > 1$ is prime if and only if $(n-1)! \equiv -1 \pmod{n}$. Unfortunately, this test is of more theoretical than practical interest because as n increases, $(n-1)!$ rapidly becomes unmanageable in size.

We would like to close this chapter with an application of Wilson's theorem to the study of quadratic congruences. [It is understood that *quadratic congruence* means a congruence of the form $ax^2 + bx + c \equiv 0 \pmod{n}$, with $a \not\equiv 0 \pmod{n}$.] This is the content of Theorem 5.5.

Theorem 5.5. The quadratic congruence $x^2 + 1 \equiv 0 \pmod{p}$, where p is an odd prime, has a solution if and only if $p \equiv 1 \pmod{4}$.

Proof. Let a be any solution of $x^2 + 1 \equiv 0 \pmod{p}$, so that $a^2 \equiv -1 \pmod{p}$. Because $p \nmid a$, the outcome of applying Fermat's theorem is

$$1 \equiv a^{p-1} \equiv (a^2)^{(p-1)/2} \equiv (-1)^{(p-1)/2} \pmod{p}$$

The possibility that $p = 4k + 3$ for some k does not arise. If it did, we would have

$$(-1)^{(p-1)/2} = (-1)^{2k+1} = -1$$

hence, $1 \equiv -1 \pmod{p}$. The net result of this is that $p \mid 2$, which is patently false. Therefore, p must be of the form $4k + 1$.

Now for the opposite direction. In the product

$$(p-1)! = 1 \cdot 2 \cdots \frac{p-1}{2} \cdot \frac{p+1}{2} \cdots (p-2)(p-1)$$

we have the congruences

$$\begin{aligned} p - 1 &\equiv -1 \pmod{p} \\ p - 2 &\equiv -2 \pmod{p} \\ &\vdots \\ \frac{p+1}{2} &\equiv -\frac{p-1}{2} \pmod{p} \end{aligned}$$

Rearranging the factors produces

$$(p-1)! \equiv 1 \cdot (-1) \cdot 2 \cdot (-2) \cdots \frac{p-1}{2} \cdot \left(-\frac{p-1}{2}\right) \pmod{p}$$

$$\equiv (-1)^{(p-1)/2} \left(1 \cdot 2 \cdots \frac{p-1}{2}\right)^2 \pmod{p}$$

because there are $(p-1)/2$ minus signs involved. It is at this point that Wilson's theorem can be brought to bear; for, $(p-1)! \equiv -1 \pmod{p}$, whence

$$-1 \equiv (-1)^{(p-1)/2} \left[\left(\frac{p-1}{2}\right)!\right]^2 \pmod{p}$$

If we assume that p is of the form $4k+1$, then $(-1)^{(p-1)/2} = 1$, leaving us with the congruence

$$-1 \equiv \left[\left(\frac{p-1}{2}\right)!\right]^2 \pmod{p}$$

The conclusion is that the integer $[(p-1)/2]!$ satisfies the quadratic congruence $x^2 + 1 \equiv 0 \pmod{p}$.

Let us take a look at an actual example, say, the case $p = 13$, which is a prime of the form $4k+1$. Here, we have $(p-1)/2 = 6$, and it is easy to see that

$$6! = 720 \equiv 5 \pmod{13}$$

and

$$5^2 + 1 = 26 \equiv 0 \pmod{13}$$

Thus, the assertion that $[((p-1)/2)!]^2 + 1 \equiv 0 \pmod{p}$ is correct for $p = 13$.

Wilson's theorem implies that there exists an infinitude of composite numbers of the form $n! + 1$. On the other hand, it is an open question whether $n! + 1$ is prime for infinitely many values of n. The only values of n in the range $1 \leq n \leq 100$ for which $n! + 1$ is known to be a prime number are $n = 1, 2, 3, 11, 27, 37, 41, 73$, and 77. Currently, the largest prime of the form $n! + 1$ is $6380! + 1$, discovered in 2000.

PROBLEMS 5.3

1. (a) Find the remainder when 15! is divided by 17.
   (b) Find the remainder when 2(26!) is divided by 29.
2. Determine whether 17 is a prime by deciding whether $16! \equiv -1 \pmod{17}$.
3. Arrange the integers $2, 3, 4, \ldots, 21$ in pairs a and b that satisfy $ab \equiv 1 \pmod{23}$.
4. Show that $18! \equiv -1 \pmod{437}$.
5. (a) Prove that an integer $n > 1$ is prime if and only if $(n-2)! \equiv 1 \pmod{n}$.
   (b) If n is a composite integer, show that $(n-1)! \equiv 0 \pmod{n}$, except when $n = 4$.
6. Given a prime number p, establish the congruence

$$(p-1)! \equiv p - 1 \pmod{1 + 2 + 3 + \cdots + (p-1)}$$

7. If p is a prime, prove that for any integer a,

$$p \mid a^p + (p-1)!a \qquad \text{and} \qquad p \mid (p-1)!a^p + a$$

[*Hint:* By Wilson's theorem, $a^p + (p-1)!a \equiv a^p - a \pmod{p}$.]

8. Find two odd primes $p \leq 13$ for which the congruence $(p-1)! \equiv -1 \pmod{p^2}$ holds.

9. Using Wilson's theorem, prove that for any odd prime p,

$$1^2 \cdot 3^2 \cdot 5^2 \cdots (p-2)^2 \equiv (-1)^{(p+1)/2} \pmod{p}$$

[*Hint:* Because $k \equiv -(p-k) \pmod{p}$, it follows that

$$2 \cdot 4 \cdot 6 \cdots (p-1) \equiv (-1)^{(p-1)/2} 1 \cdot 3 \cdot 5 \cdots (p-2) \pmod{p}.]$$

10. (a) For a prime p of the form $4k+3$, prove that either

$$\left(\frac{p-1}{2}\right)! \equiv 1 \pmod{p} \qquad \text{or} \qquad \left(\frac{p-1}{2}\right)! \equiv -1 \pmod{p}$$

hence, $[(p-1)/2]!$ satisfies the quadratic congruence $x^2 \equiv 1 \pmod{p}$.

(b) Use part (a) to show that if $p = 4k+3$ is prime, then the product of all the even integers less than p is congruent modulo p to either 1 or -1.
[*Hint:* Fermat's theorem implies that $2^{(p-1)/2} \equiv \pm 1 \pmod{p}$.]

11. Apply Theorem 5.5 to obtain two solutions to each of the quadratic congruences $x^2 \equiv -1 \pmod{29}$ and $x^2 \equiv -1 \pmod{37}$.

12. Show that if $p = 4k+3$ is prime and $a^2 + b^2 \equiv 0 \pmod{p}$, then $a \equiv b \equiv 0 \pmod{p}$.
[*Hint:* If $a \not\equiv 0 \pmod{p}$, then there exists an integer c such that $ac \equiv 1 \pmod{p}$; use this fact to contradict Theorem 5.5.]

13. Supply any missing details in the following proof of the irrationality of $\sqrt{2}$: Suppose $\sqrt{2} = a/b$, with $\gcd(a, b) = 1$. Then $a^2 = 2b^2$, so that $a^2 + b^2 = 3b^2$. But $3 \mid (a^2 + b^2)$ implies that $3 \mid a$ and $3 \mid b$, a contradiction.

14. Prove that the odd prime divisors of the integer $n^2 + 1$ are of the form $4k+1$.
[*Hint:* Theorem 5.5.]

15. Verify that $4(29!) + 5!$ is divisible by 31.

16. For a prime p and $0 \leq k \leq p-1$, show that $k!(p-k-1)! \equiv (-1)^{k+1} \pmod{p}$.

17. If p and q are distinct primes, prove that for any integer a,

$$pq \mid a^{pq} - a^p - a^q + a$$

18. Prove that if p and $p+2$ are a pair of twin primes, then

$$4((p-1)! + 1) + p \equiv 0 \pmod{p(p+2)}$$

5.4 THE FERMAT-KRAITCHIK FACTORIZATION METHOD

In a fragment of a letter, written in all probability to Father Marin Mersenne in 1643, Fermat described a technique of his for factoring large numbers. This represented the first real improvement over the classical method of attempting to find a factor of n by dividing by all primes not exceeding $\sqrt{n}$. Fermat's factorization scheme has at its heart the observation that the search for factors of an odd integer n (because powers of 2 are easily recognizable and may be removed at the outset, there is no loss in assuming that n is odd) is equivalent to obtaining integral solutions x and y of the equation

$$n = x^2 - y^2$$